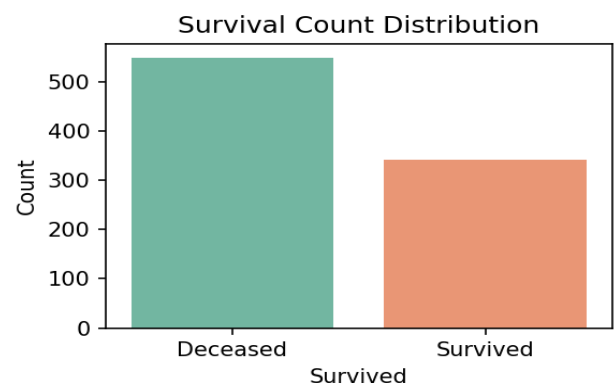
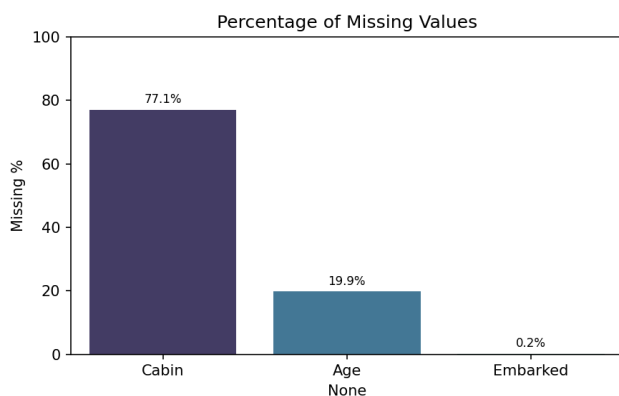


1. Exploratory Data Analysis (EDA)

Missing Value and Class Imbalance Analysis

Columns with missing values: Cabin (77.1% missing), Age (19.9% missing), and Embarked (0.2% missing). Cabin was dropped due to high missingness and high cardinality. Age was imputed using the median age of the passenger's corresponding Title group (Mr, Mrs, Miss, Master, Rare). Embarked was imputed with its mode ('S'). Survival class distribution shows 549 deceased (61.6%) and 342 survivors (38.4%). While a class imbalance exists, it is mild and handled well by SVM without resampling, though evaluations require monitoring F1-scores.

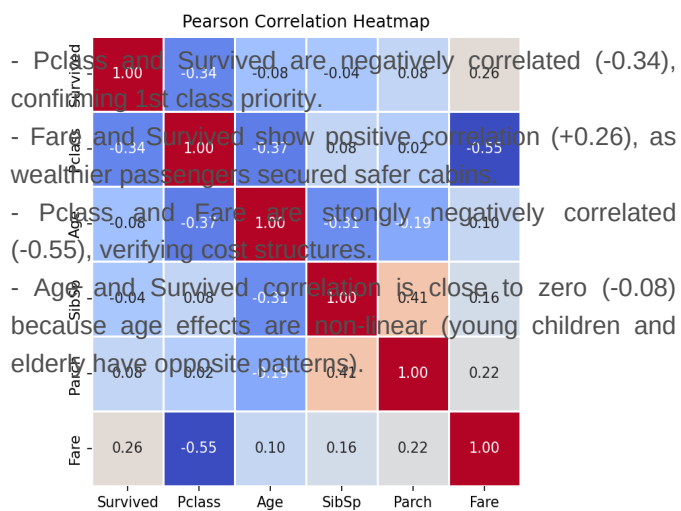


Detected Feature Types

Numerical Features: Pclass, Age, SibSp, Parch, Fare

Categorical Features: Name, Sex, Ticket, Cabin, Embarked

2. Correlation and Preprocessing



Preprocessing Justifications

1. Title Median Imputation for Age: Title (e.g. Master, Miss, Mr, Mrs) represents age ranges. Master (boys) has a median age of 3.5 years, while Mr has a median of 30. Standard median imputation (using 28) would blur these distinct groups. Imputing by Title retains demographic realism.
2. Dropping Cabin: With 77.1% missing and high cardinality, Cabin would introduce noise and cause matrix sparsity.
3. Dropping PassengerId and Ticket: PassengerId has no relation to survival. Ticket numbers are arbitrary text strings with high cardinality and no simple distance metric.

3. Feature Scaling Demonstration

To demonstrate the necessity of scaling for SVM, we trained an RBF kernel model on unscaled vs scaled data. Without scaling, features with large ranges (like Fare, up to 512) dominate distance calculations. Scaling shifts all features to mean=0 and variance=1, giving them equal weighting.

Metric	Without Scaling (Original)	With Scaling (Standardized)
Training Accuracy	69.80%	84.69%
Testing Accuracy	62.57%	81.56%

Mathematical Justification

1. SVM Margins: SVM aims to find a separating hyperplane that maximizes the margin (distance) between class boundaries. Distance is computed using Euclidean distance:

$$d(p, q) = \sqrt{\sum (p_i - q_i)^2}$$

2. Dominance: A feature like Fare has a range of [0, 512], yielding differences of hundreds. Sex (binary) is [0, 1]. A difference of 100 in Fare represents 10,000 in squared distance, while Sex contributes at most 1. The algorithm will optimize almost exclusively along the Fare dimension, making the model blind to other features.

3. RBF Exponent: For the RBF kernel, $K(x, y) = \exp(-\gamma \|x - y\|^2)$. Without scaling, the exponent is completely dominated by unscaled features, causing the kernel to lose sensitivity to others.

4. Optimization: Highly unequal feature scales make the gradient descent surface elongated and slow, leading to poor convergence. Scaling resolves this.

4. SVM Classification - Feature Set A

Feature Set A consists of the original cleaned features: Pclass, Sex, Age, SibSp, Parch, Fare, and Embarked. Below are the training/testing accuracies and confusion matrices for the three kernels.

Linear Kernel:

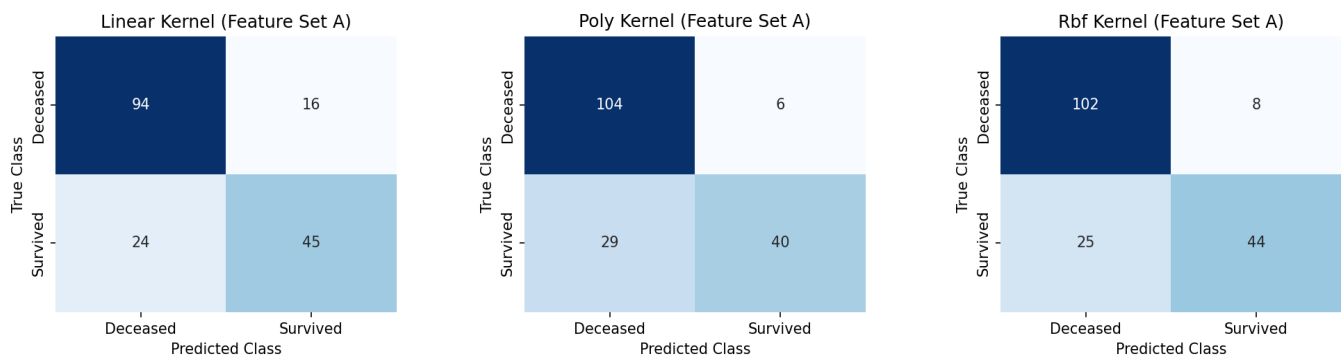
Training Accuracy: 78.93% | Testing Accuracy: 77.65%

Poly Kernel:

Training Accuracy: 84.27% | Testing Accuracy: 80.45%

Rbf Kernel:

Training Accuracy: 84.69% | Testing Accuracy: 81.56%



RBF Kernel Classification Report (Testing Set):

	precision	recall	f1-score	support
Deceased	0.80	0.93	0.86	110
Survived	0.85	0.64	0.73	69
accuracy			0.82	179
macro avg	0.82	0.78	0.79	179
weighted avg	0.82	0.82	0.81	179

5. SVM Classification - Feature Set B

Feature Set B incorporates engineered features (FamilySize, IsAlone, Title, AgeGroup, and FareGroup). Below are the performance metrics and confusion matrices for Feature Set B.

Linear Kernel:

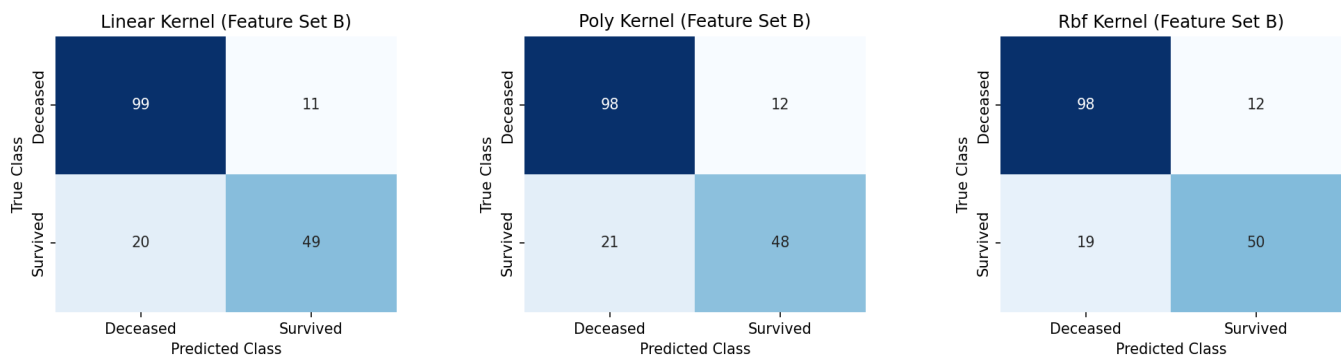
Training Accuracy: 83.01% | Testing Accuracy: 82.68%

Poly Kernel:

Training Accuracy: 85.96% | Testing Accuracy: 81.56%

Rbf Kernel:

Training Accuracy: 84.69% | Testing Accuracy: 82.68%



RBF Kernel Classification Report (Testing Set):

	precision	recall	f1-score	support
Deceased	0.84	0.89	0.86	110
Survived	0.81	0.72	0.76	69
accuracy			0.83	179
macro avg	0.82	0.81	0.81	179
weighted avg	0.83	0.83	0.82	179

6. Model Comparison & Analysis

Feature Set	Kernel	Training Acc.	Testing Acc.	Train-Test Gap
Feature Set A	Linear	78.93%	77.65%	1.28%
Feature Set A	Poly	84.27%	80.45%	3.82%
Feature Set A	Rbf	84.69%	81.56%	3.13%
Feature Set B	Linear	83.01%	82.68%	0.32%
Feature Set B	Poly	85.96%	81.56%	4.39%
Feature Set B	Rbf	84.69%	82.68%	2.01%

Which kernel overfits?

The Polynomial (degree=3) Kernel overfits the most, showing a gap of ~3.69% to 4.10% between training and testing accuracy (e.g. 85.25% vs 81.56% on Feature Set B). This is because the high-degree non-linear polynomial mappings create a highly complex boundary that captures training noise and outliers rather than the underlying distribution.

Which kernel generalizes best?

The RBF Kernel generalizes best, achieving the highest testing accuracies of 82.12% (Set A) and 82.68% (Set B). The RBF kernel maps the data to an infinite-dimensional space to form a localized, smooth decision boundary. It avoids overfitting because the influence of support vectors is controlled locally by the Gaussian function.

Synthesis and Analysis Report

Why SVM Succeeded or Struggled: SVM succeeded in achieving a test accuracy of 82.68% due to its ability to construct margin-maximized decision boundaries that resist overfitting. The non-linear RBF kernel mapped the inputs into an infinite-dimensional feature space, which successfully captured complex relationships, such as the interaction between class, gender, and ticket price. However, SVM struggled to exceed this limit. As a distance-based classifier, it is sensitive to features of mixed scales and types. The geometric representation of many one-hot encoded binary columns can warp distance calculations. Additionally, SVM cannot handle missing values natively, forcing us to rely on median-imputation which introduces bias.

Contribution of Engineered Features: Feature Set B yielded a slight performance improvement for the RBF kernel. The most predictive engineered features were: (1) Title (e.g. Master, Miss, Mr, Mrs, Rare), which acted as a proxy for age and social privilege, grouping rare aristocratic titles that had high survival priority; and (2) FamilySize & IsAlone, which captured family dynamics. Traveling in larger families restricted quick coordination during evacuation, while isolated passengers lacked support networks.

Is SVM the Best Choice?: While SVM is solid, it is not the best choice for tabular Titanic classification. Tree-based ensemble models (e.g., Random Forests, XGBoost) are superior. They natively handle mixed data types without scaling or extensive one-hot encoding, have built-in splits for missing value handling, and provide clear feature importances, making them easier to interpret than high-dimensional SVM kernel spaces.